

Generative AI for Short-Video Feed Recommendation on KuaiRec: A Proxy Study for In-Feed Ad Optimization

Short-form video platforms such as Instagram Reels, TikTok, and Kuaishou are now central to how people consume digital content. In these feeds, ads are shown as short videos mixed in between normal clips, which makes the feed a key channel for both user engagement and monetization (Tran, 2024). This raises a core challenge: how to design recommendation policies that select the “next video” or “next ad” from a large pool of candidates in a way that maximizes engagement, supports long-term user satisfaction, and avoids over-exposing only the most popular content. The KuaiRec dataset helps study this problem by providing a fully observed user–item matrix in which 1,411 users have watched nearly all 3,327 videos, together with a larger but sparser matrix over 7,176 users and 10,728 items. This structure enables realistic offline simulators for interactive recommendation. Prior work on KuaiRec shows that data density and exposure bias have a strong impact on the reliability of offline evaluation and highlights the importance of fully observed benchmarks when analyzing recommendation algorithms [1].

Building on KuaiRec, the DORL (Debiased Model-based Offline Reinforcement Learning) framework treats recommendation as an offline reinforcement learning problem. At a high level, it uses viewing time as a reward signal and adds a debiasing term to reduce the “Matthew effect,” where already popular categories are recommended too often. Results show that this approach improves long-term reward and reduces popularity domination compared to several standard offline RL baselines [2]. This line of work shows that KuaiRec can be used not only for static recommendation, but also for studying long-horizon, fairness-aware policies in short-video environments. However, both the original KuaiRec paper and the DORL framework mainly rely on ID-based and hand-crafted user and item features. They do not use large language models or generative representation learning, and they treat all items as normal videos without explicitly modeling in-feed ads or auction signals.

To address these gaps, our study project explores how Generative AI, in particular large language models and generative representation learning, can improve short-video feed recommendation on KuaiRec as a proxy for in-feed ad optimization (ads shown inside the scrolling video feed). First, we build LLM-based item representations using KuaiRec’s textual metadata (video captions, topic tags, and hierarchical category labels). Instead of representing

each video mainly as an anonymous ID with simple one-hot features, we prompt an LLM to generate semantic descriptions and embeddings for each video based on its metadata. Then use these LLM-derived embeddings as item features in our recommendation or offline RL models and compare their performance to standard ID-based embeddings from prior work.

Second, we extend this idea to user-side generative features. We summarize each user's interaction history (the sequence of videos they watched and the associated metadata) into a short, natural-language profile using an LLM, and then encode this profile into a continuous embedding. The goal is to capture higher-level user interests—such as preferred themes, creators, or pacing styles—that are difficult to represent with simple numeric features alone. By incorporating both generative item and user representations into KuaiRec-based environments, we aim to test whether generative modelling can (a) increase engagement-oriented metrics such as cumulative reward and interaction length, and (b) reduce over-reliance on a small set of highly popular content categories.

Although KuaiRec does not provide explicit advertisement or auction signals, its short-video feed structure is very similar to in-feed advertising: in both cases, the system must choose the next video-like item from a candidate set to keep the user engaged. We therefore view this study as an initial proof-of-concept for applying generative AI to ad-like recommendation scenarios. Insights from LLM-based representations and generative RL on KuaiRec could guide future work on real advertising logs that additionally model bidding, budget constraints, and conversion outcomes, helping to connect generative recommendation methods to practical in-feed ad delivery systems.

[1] Gao, C., Li, S., Lei, W., Chen, J., Li, B., Jiang, P., ... & Chua, T. S. (2022, October). *KuaiRec: A fully-observed dataset and insights for evaluating recommender systems*. In Proceedings of the 31st ACM International Conference on Information & Knowledge Management (pp. 540–550).

[2] Gao, C., Huang, K., Chen, J., Zhang, Y., Li, B., Jiang, P., ... & He, X. (2023, July). *Alleviating Matthew effect of offline reinforcement learning in interactive recommendation*. In Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval (pp. 238–248).

[3] Tran, A. (2024). *Optimization of social media video in visual art self-marketing*. (Bachelor's thesis, Tampere University of Applied Sciences)